

Peter Dietrich
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February 14, 2020
NRC-20-0007

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2019-006

Pursuant to 10 CFR 50.73(a)(2)(v)(C), DTE Electric Company (DTE) is submitting LER No. 2019-006, Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Jason R. Haas, Manager – Nuclear Licensing, at (734) 586-1769.

Sincerely,

A handwritten signature in black ink, appearing to be "P. Dietrich", written over a large, stylized circular flourish.

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2019-006, Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III
Michigan Department of Environment, Great Lakes, and Energy

**Enclosure to
NRC-20-0007**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2019-006, Secondary Containment Pressure Exceeded
Technical Specification Due to Reactor Building HVAC Damper Malfunction**

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Fermi 2	2. Docket Number 05000 341	3. Page 1 OF 4
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4. Title Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
12	18	2019	2019	006	00	02	14	2020	N/A	05000
									N/A	05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER	
Licensee Contact Fermi 2 / Jason R. Haas – Manager, Nuclear Licensing	Telephone Number (Include Area Code) (734) 586-1769

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	VA	75	B045	Y	N/A	N/A	N/A	N/A	N/A
14. Supplemental Report Expected					15. Expected Submission Date				
☐ Yes (If yes, complete 15. Expected Submission Date)					☒ No				
					Month Day Year				

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On December 18, 2019, at 0908 EST, with the East and Center Reactor Building Heating Ventilation and Air Conditioning (RBHVAC) trains in service, secondary containment (SC) pressure degraded to the point where the Technical Specification (TS) for SC pressure was exceeded for approximately 4 minutes. The maximum SC pressure recorded during that time was 0.064 inches of vacuum water gauge. SC pressure was restored to within TS limits by manually starting the Division 1 Standby Gas Treatment System. There were no safety consequences or radiological releases associated with this event. The cause of this momentary loss of SC TS requirements was determined to be a failure in the Center RBHVAC exhaust fan inlet damper positioner. For corrective actions, Fermi 2 has upgraded two of the three RBHVAC exhaust inlet damper positioners and scheduled an upgrade replacement of the third.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2	05000-341	YEAR 2019	SEQUENTIAL NUMBER 006	REV NO. 00

NARRATIVE**INITIAL PLANT CONDITIONS**

Mode – 1
Reactor Power – 100%

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On December 18, 2019, at 0908 EST, with the unit in Mode 1 at 100 percent power and the East and Center Reactor Building Heating Ventilation and Air Conditioning (RBHVAC) [[VA]] trains in service, an alarm indicated that secondary containment (SC) [[NH]] pressure had degraded to the point where the Technical Specification (TS) requirement for SC pressure was exceeded and SC was declared inoperable. SC pressure was returned to within the TS operability limit of greater than or equal to 0.125 inches of vacuum water gauge per TS Surveillance Requirement (SR) 3.6.4.1.1 by manually starting Division 1 of the Standby Gas Treatment System (SGTS) [[BH]]. SC was declared OPERABLE on December 18, 2019, at 0912 EST. The duration of the TS not being met was approximately 4 minutes and the maximum SC pressure observed during the event was approximately 0.064 inches of vacuum water gauge. A modulating damper [[DMP]] associated with the Center train of RBHVAC was identified as not properly controlling.

An 8-hour event notification (EN 54452) was made to the NRC based on meeting the reporting criteria of Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(3)(v)(C) as an event or condition that could have prevented the fulfillment of a safety function needed to control the release of radioactive material. This Licensee Event Report (LER) is being made under the corresponding requirement in 10 CFR 50.73(a)(2)(v)(C).

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

The specified safety function of the SC is to contain, dilute, and hold up fission products that may leak from primary containment following a Design Basis Accident (DBA). In conjunction with operation of the SGTS and closure of certain valves [[V]] whose lines penetrate the SC, the SC is designed to reduce the activity level of the fission products prior to release to the environment and to isolate and contain fission products that are released during certain operations that take place inside primary containment, when primary containment is not required to be OPERABLE, or that take place outside primary containment. It is possible for the pressure in the control volume to rise relative to the environmental pressure (e.g., due to pump [[P]] and motor [[MO]] heat load additions). To prevent ground level exfiltration while allowing the SC to be designed as a conventional structure, the SC requires support systems to maintain the control volume pressure at less than the external pressure. For the SC to be considered OPERABLE, it must have adequate leak tightness to ensure that the required vacuum can be established and maintained.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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Fermi 2	05000-	YEAR	SEQUENTIAL NUMBER	REV NO.
	341	2019	006	00

NARRATIVE

The RBHVAC system does not have any safety related functions except isolating SC following a DBA, by closing separate safety-related dampers. These safety related isolation dampers were not impacted. RBHVAC malfunctions can result in Reactor Building pressure perturbations, however the system will trip on a Loss of Coolant Accident (LOCA) or other signals. When the non-safety related RBHVAC trips on any of these signals, the safety related systems will function. Specifically, the Reactor Building SC isolation dampers close and SGTS automatically initiates to draw down SC.

In Chapters 6 and 15 of the Updated Final Safety Analysis Report (UFSAR), RBHVAC is assumed lost at the onset of a LOCA concurrent with a Loss of Offsite Power. As a result, calculations show that the SC would be pressurized until the SGTS restores vacuum. The structural integrity (i.e., leak tightness) of the SC was re-confirmed on December 18 when SC vacuum was restored to greater than 0.125 inches of vacuum water gauge by manually starting Division 1 SGTS.

If the DBA LOCA concurrent with a Loss of Offsite Power had occurred during the time when the SC pressure TS limit was exceeded, the SC was sufficiently leak tight such that the SGTS would still have established and maintained vacuum greater than the TS required value without RBHVAC in service.

The radiological consequences of the DBA LOCA for SC contained in Chapter 15 of the UFSAR result in doses that are below 10 CFR 50.67. The SC is assumed to be at a pressure of 0.0 inches of vacuum water gauge at the onset of the LOCA. For this particular event, had the DBA LOCA for SC actually occurred there would be no increase in magnitude of radiological dose because the highest recorded pressure of 0.064 inches of vacuum water gauge was bounded by the assumed SC pressure of 0.0 inches of vacuum water gauge.

CAUSE OF THE EVENT

The failure mechanism was assembly looseness in the Center RBHVAC exhaust fan inlet damper positioner [[75]] (Bailey model AV-1). Due to the normal modulating of the positioner, the cam follower bearing clip came off, causing the cam to become dislodged which then cut through the bearing shaft. The failure of the positioner did not allow the damper to modulate, resulting in the increased SC pressure. The failure of the positioner was caused by an inadequate vendor design. Contributing to the cause, external Operating Experience was not adequately applied to prevent this event.

CORRECTIVE ACTIONS

The immediate corrective action was to manually start Division 1 of the SGTS to restore SC vacuum to within the TS operability limit. The RBHVAC system was removed from service for troubleshooting on 12/18/2019 at 0927. RBHVAC was restarted utilizing the East RBHVAC train on 12/18/2019 at 2258.

The Center RBHVAC exhaust fan damper positioner is scheduled for replacement with a more robust cam follower assembly in accordance with the Fermi 2 corrective action program. Until the positioner is replaced, the Center RBHVAC is only available for emergency use. The East RBHVAC exhaust fan inlet damper positioner was upgraded on 11/26/19 and the West was upgraded on 12/27/19.

Engineering performed an extent of cause to find the remaining inadequate Bailey AV-1 positioners installed in the plant and in stock. Corrective actions have been initiated.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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	341	2019	006	00

NARRATIVE

PREVIOUS OCCURRENCES

Events involving loss of SC due to the issues with the RBHVAC system have been reported in past, including the following recent LERs:

LER 2015-001 involved the loss of SC function due to an RBHVAC system trip caused by a valid actuation of a freeze protection device.

LER 2015-004 involved the loss of SC function due to reverse rotation of the RBHVAC Center exhaust fan during post-maintenance testing caused by reversed electrical leads.

LER 2015-005-01 involved the loss of SC function due to setpoint drift of the RBHVAC supply damper time delay relay resulting in the dampers moving out of sequence.

LER 2016-005 involved the loss of SC function due to the combined effect of high winds during the RBHVAC startup sequence.

LER 2018-001 involved the loss of SC function due to an RBHVAC exhaust fan modulating damper failing to full open due to a mechanical failure in the damper linkage.

LER 2019-005 involved the loss of SC function due to an RBHVAC exhaust fan discharge damper failing to close due to a sticking time delay relay.

None of the referenced occurrences involved the damper positioner, thus there is not a similar underlying concern or reason to this event, such as the same root cause, failure, or sequence of events.